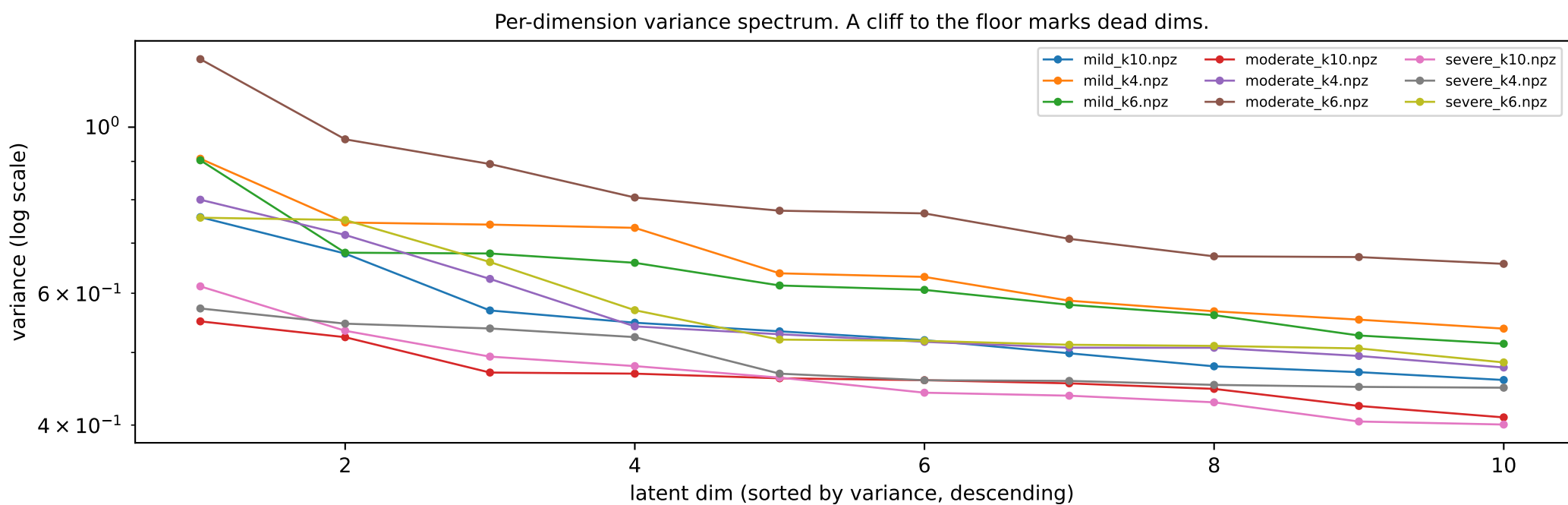
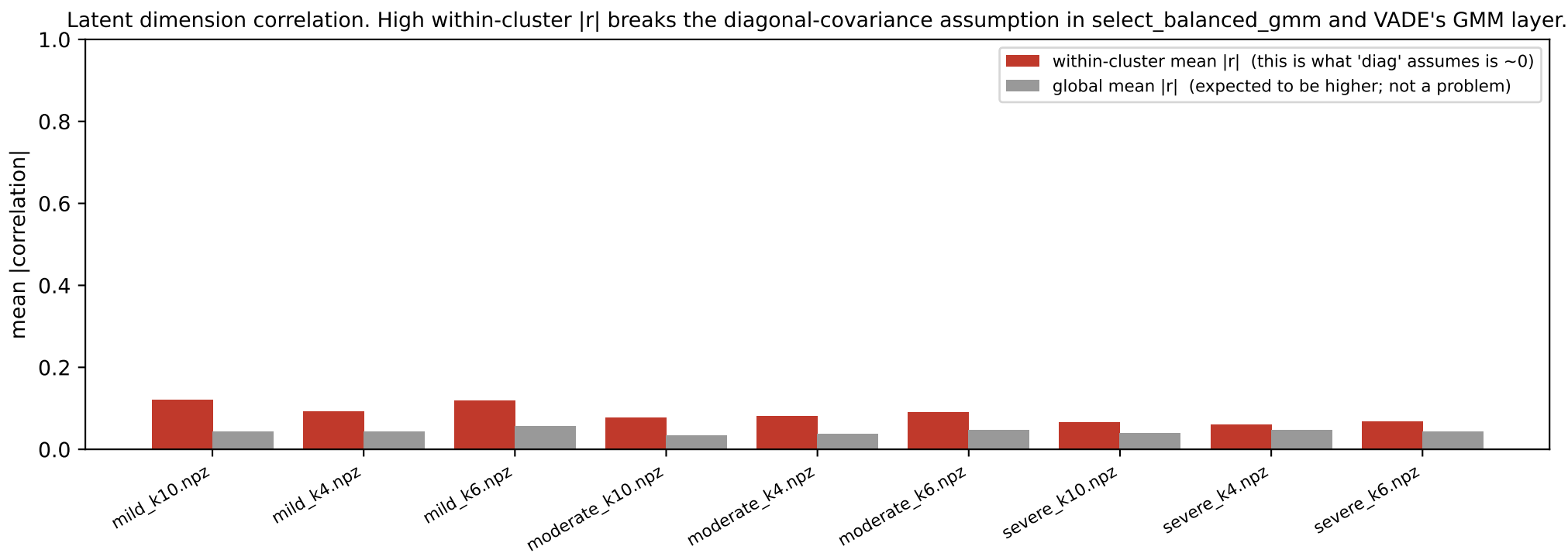
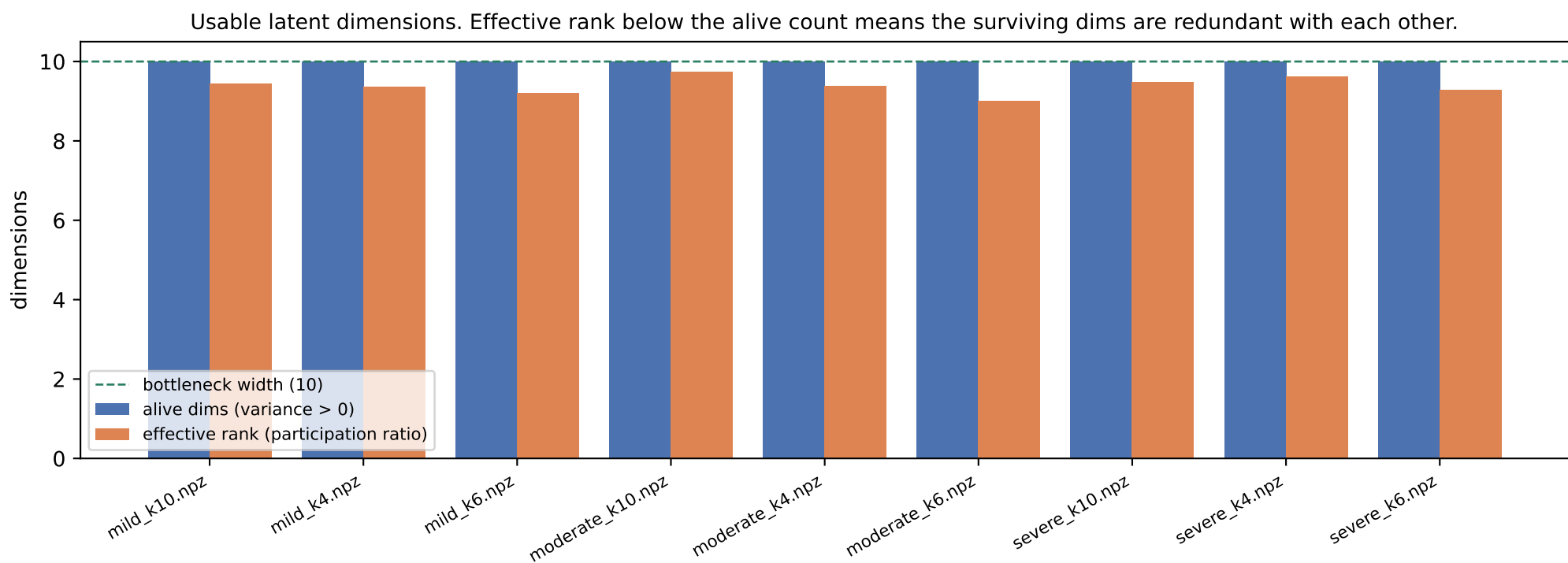
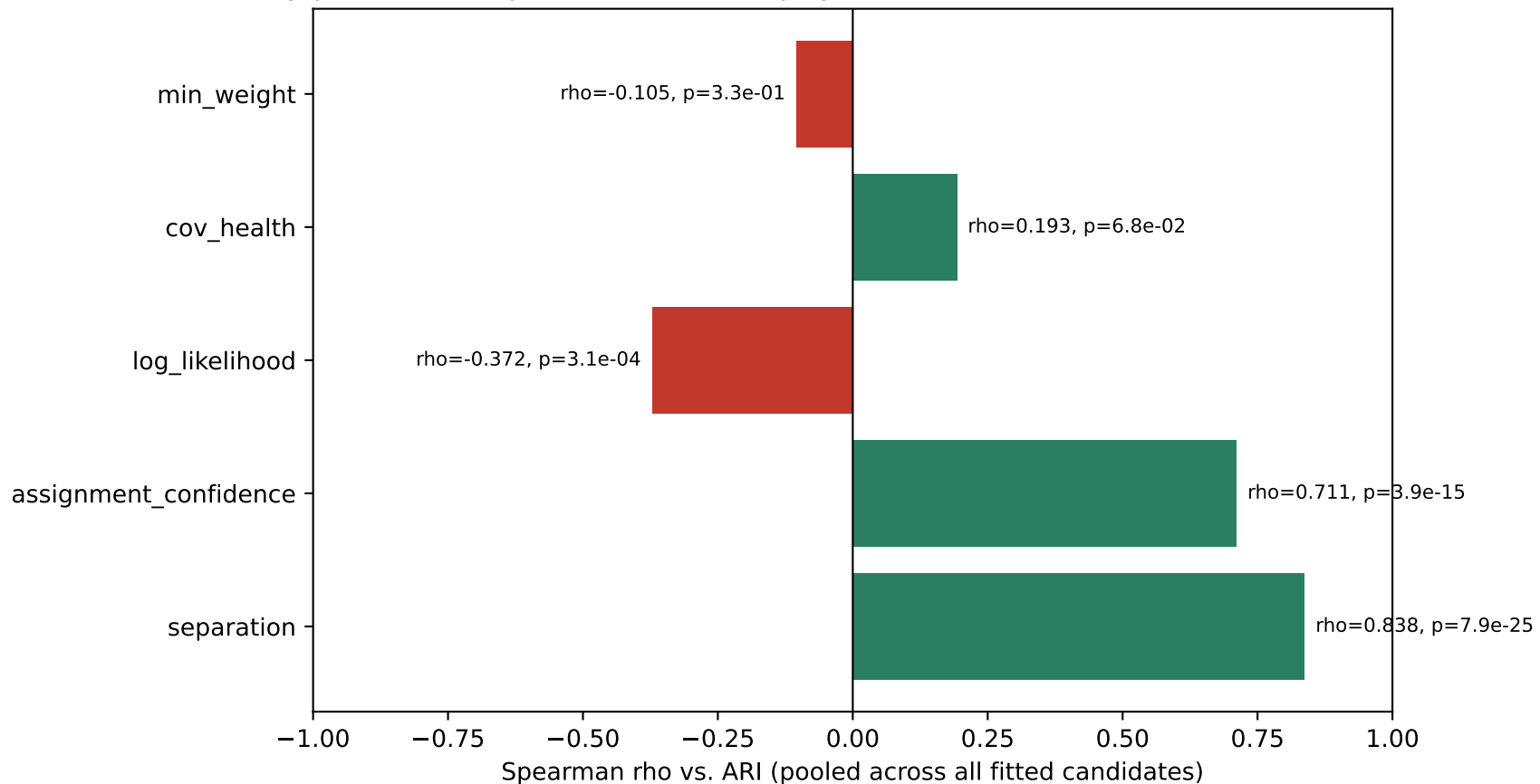


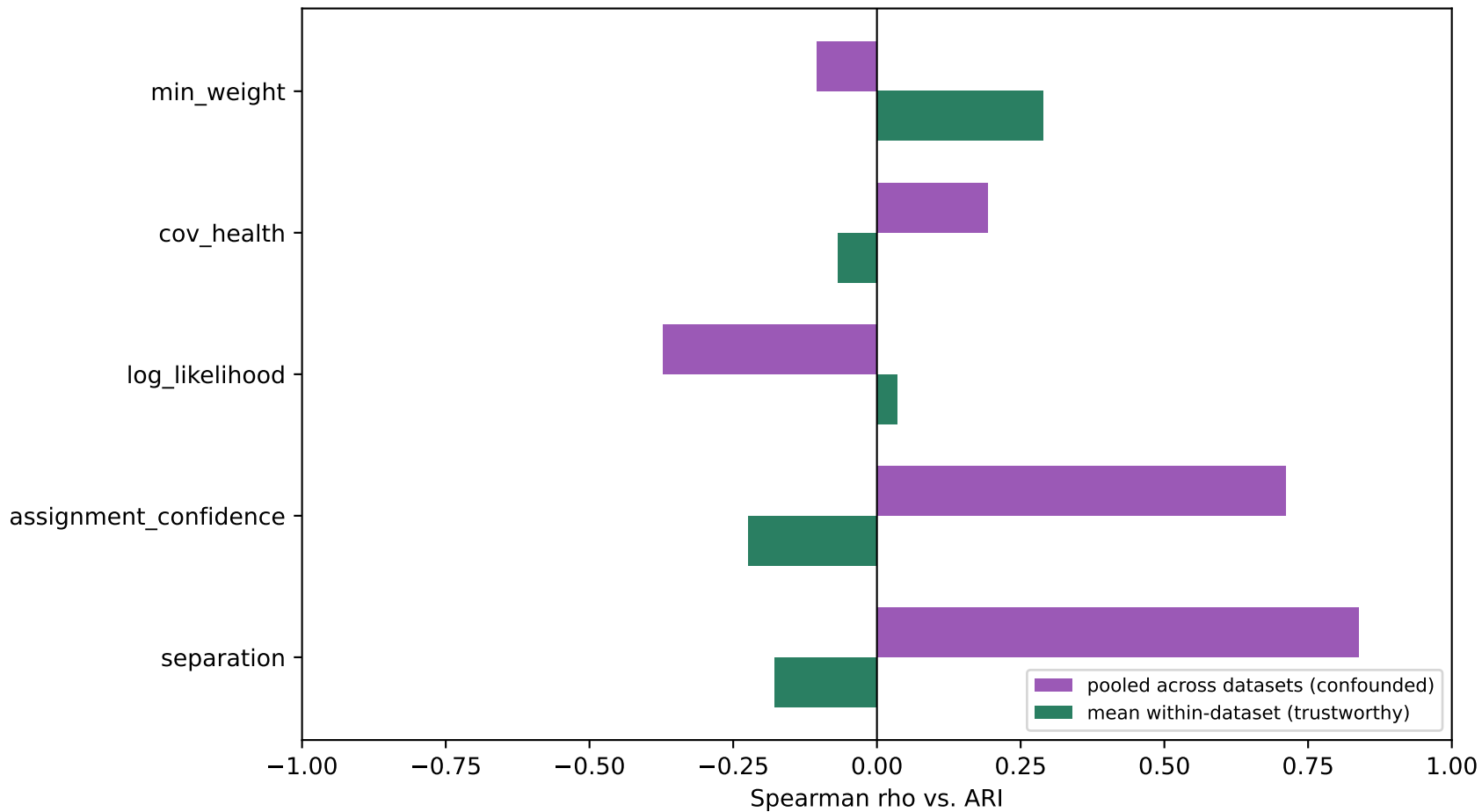
Latent-space health (latent_activation=None)



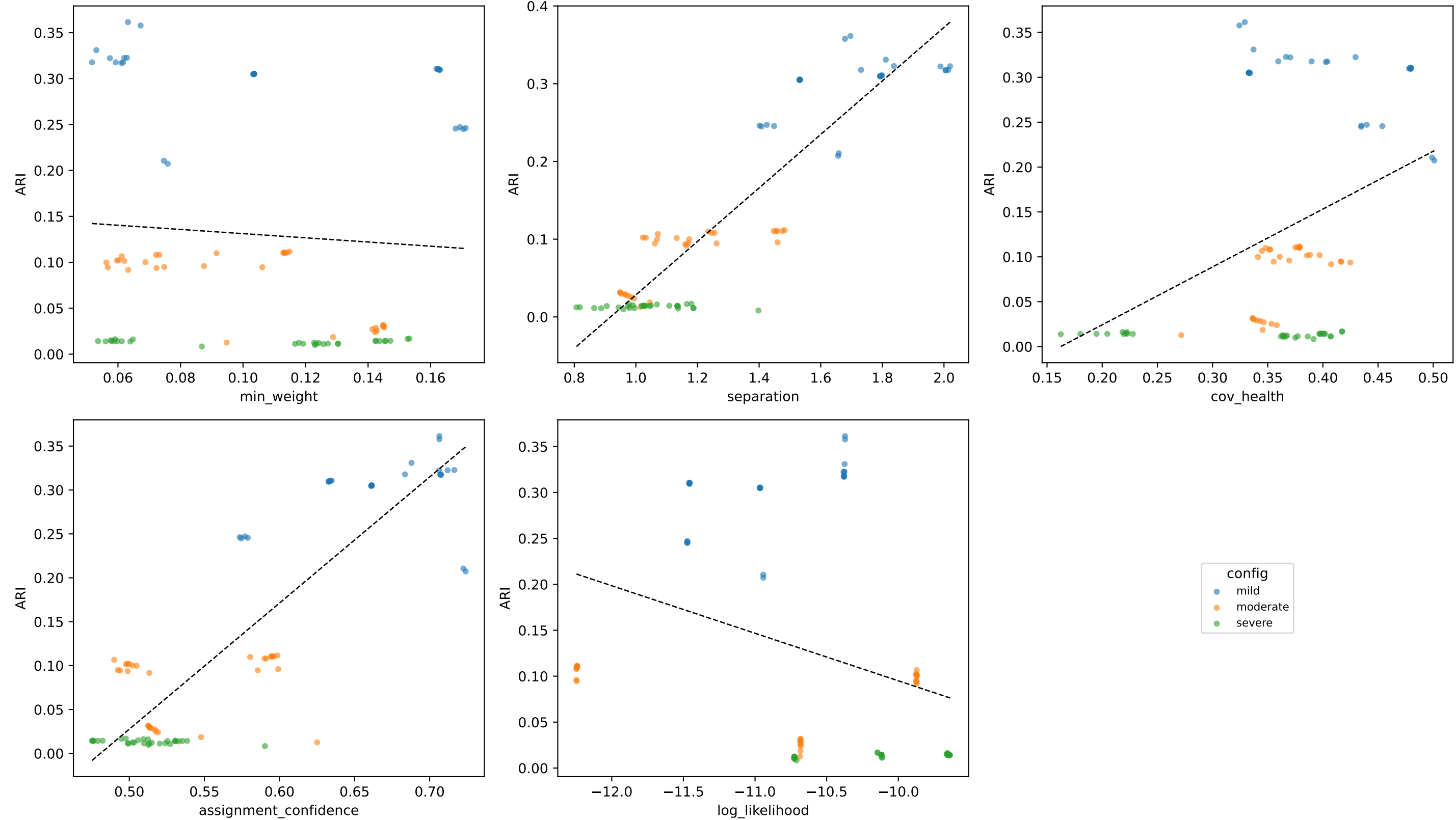
h raw metrics actually predict ARI? (pooled -- see next page: this inflates/deflates metrics via cross-dataset



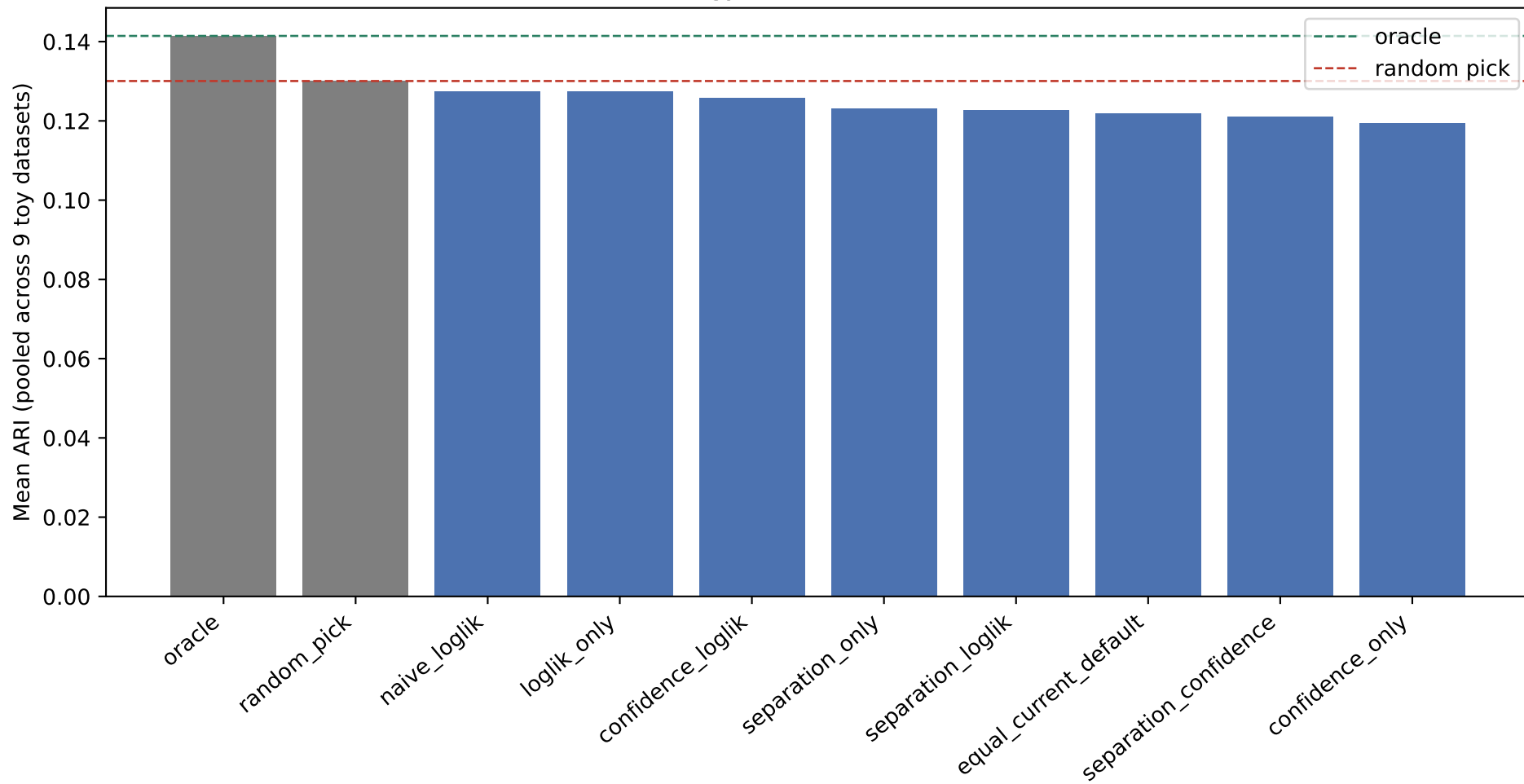
Pooled correlation can overstate (or invert) predictive power --
within-dataset rho is the number to trust



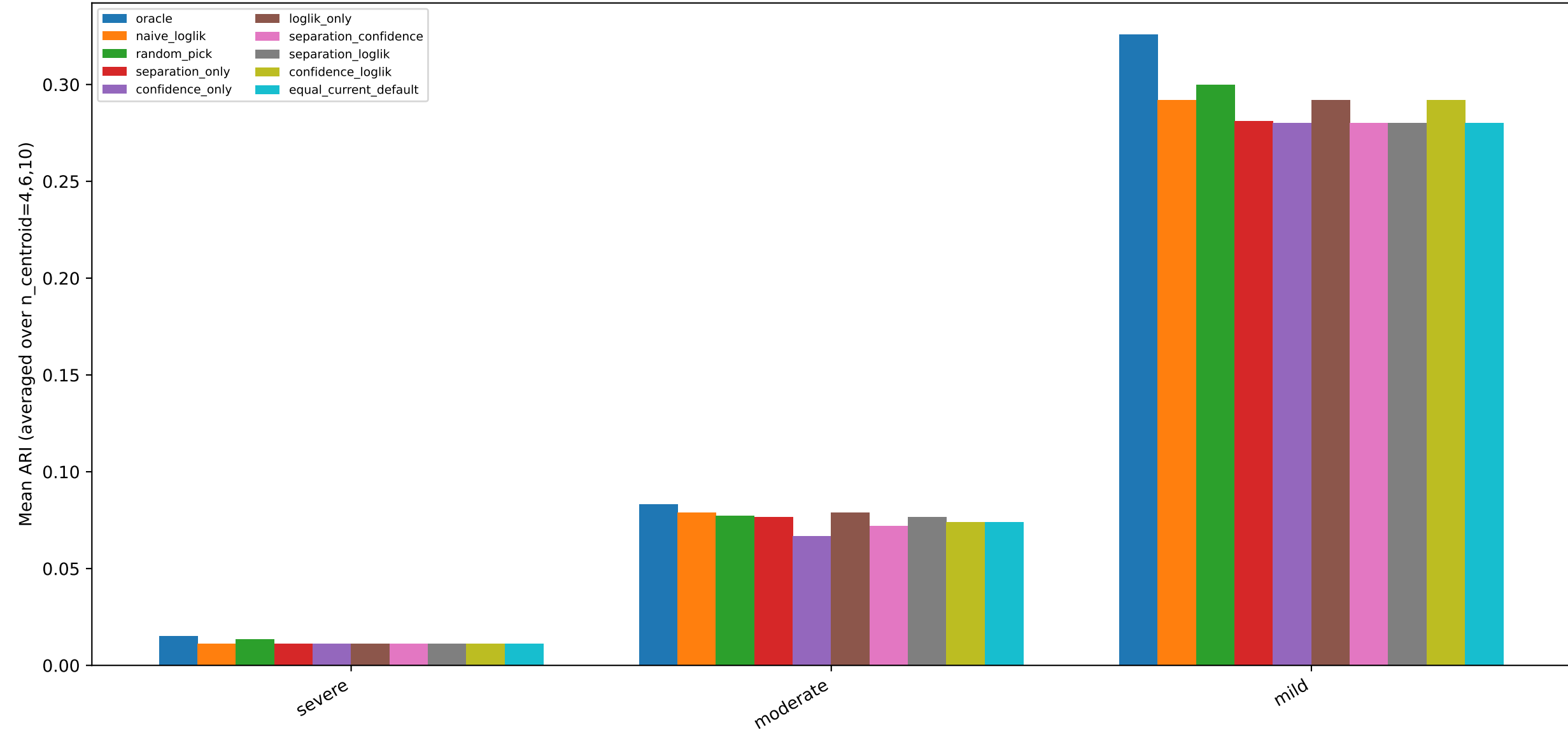
Metric value vs. ARI, per candidate (90 candidates, colored by stress config)



Selection strategy vs. oracle / random baselines



Strategy ARI by stress config — where does the metric choice matter?



Metric selection summary

Metric → ARI correlation ranking (pooled, $|\rho|$ descending):

separation	$\rho=+0.838$	$p=7.9e-25$
assignment_confidence	$\rho=+0.711$	$p=3.9e-15$
log_likelihood	$\rho=-0.372$	$p=3.1e-04$
cov_health	$\rho=+0.193$	$p=6.8e-02$
min_weight	$\rho=-0.105$	$p=3.3e-01$

Metric → ARI correlation ranking (WITHIN-DATASET mean, $|\rho|$ descending -- trust this one):

min_weight	mean_rho=+0.289	(9/9 datasets)
assignment_confidence	mean_rho=-0.223	(9/9 datasets)
separation	mean_rho=-0.179	(9/9 datasets)
cov_health	mean_rho=-0.067	(9/9 datasets)
log_likelihood	mean_rho=+0.036	(9/9 datasets)

Top 3 selection strategies by pooled mean ARI:

oracle	mean ARI=0.1414
random_pick	mean ARI=0.1301
naive_loglik	mean ARI=0.1274

oracle	mean ARI=0.1414	(upper bound)
random_pick	mean ARI=0.1301	(lower bound)

AE pretraining loss vs. epoch, per dataset

